Part III Artificial Lift Methods

Most oil reservoirs are of the volumetric type where the driving mechanism is the expansion of solution gas when reservoir pressure declines because of fluid production. Oil reservoirs will eventually not be able to produce fluids at economical rates unless natural driving mechanisms (e.g., aquifer and/or gas cap) or pressure maintenance mechanisms (e.g., water flooding or gas injection) are present to maintain reservoir energy. The only way to obtain a high production rate of a well is to increase production pressure drawdown by reducing the bottom-hole pressure with artificial lift methods.

Approximately 50% of wells worldwide need artificial lift systems. The commonly used artificial lift methods include the following:

- Sucker rod pumping
- Gas lift
- Electrical submersible pumping
- Hydraulic piston pumping
- Hydraulic jet pumping
- Plunger lift
- Progressing cavity pumping

Each method has applications for which it is the optimum installation. Proper selection of an artificial lift method for a given production system (reservoir and fluid properties, wellbore configuration, and surface facility restraints) requires a thorough understanding of the system. Economics analysis is always performed. Relative advantages and disadvantages of artificial lift systems are discussed in the beginning of each chapter in this part of this book. The chapters in this part provide production engineers with fundamentals of sucker rod pumping and gas lifts, as well as an introduction to other artificial lift systems. The following three chapters are included in this part of the book:

Chapter 12: Sucker Rod Pumping
Chapter 13: Gas Lift
Chapter 14: Other Artificial Lift Methods